

Catalyst Academy

Complete Curriculum

Background & Problem

The [CDC's Epidemiology and Public Health Science: Core Competencies for High School Students \(EPHS\)](#) framework is an important step toward introducing secondary students to public health, but it was developed in 2015 before many of the most pressing public health priorities of today were widely recognized or integrated into education. As we look to design a high school public health educational program, we evaluated the EPHS competencies and identified key limitations that inhibit their effectiveness as a standalone foundation for modern public health education.

- Limited reflection of 21st century public health priorities
 - The EPHS framework lacks adequate focus on health equity, systemic racism, climate change, digital transformation, global health, technology, law, policy, behavioral science, economics, and ethics to name a few, all of which are now central to both education and practice in public health.
 - It does not address how students can engage as active change makers in policy, advocacy, or digital health innovation, all key pathways for today's youth interested in social impact.
- Minimal Emphasis on Community Engagement, Ethics, and Wellness
 - The EPHS model is largely technical and content focused, with limited attention to the emotional intelligence, cultural competence, civic participation, and well being that are essential for those entering professions like healthcare and public health.
 - Mentorship, identity exploration, and holistic development are largely absent from EPHS, despite being critical factors in students' engagement, belonging, and career motivation.
- EPHS lacks inspiration for Public Health
 - For many high school students, this is their first exposure to public health. An effective introduction should not only teach the science but also ignite curiosity, deepen purpose, and empower students to see themselves as part of the solution.



- EPHS does not provide the narrative context or real world relevance that helps students connect personal values to public health action.
- Overemphasis on Scientific Research Methods
 - 2 of the 4 domains of the EPHS Core Competencies focus on study design and statistical modeling, concepts that can be redundant with a high school student's other quantitative coursework and are not necessarily relevant for this early stage in a student's public health career.
 - EPHS Core Competencies Domains 2 & 3 are focused on data collection and data analysis, respectively.
 - At this stage in the student's public health career, the student does not yet have enough fundamental public health knowledge to conduct studies nor collect & analyze empirical data. Teaching these skills in depth is not relevant to where the student is in their public health education. It is unlikely that a student will be conducting, for example, clinical trials (as mentioned in the EPHS core competencies) at this stage in their public health career.
 - A public health course for high school students should aim to provide an overview of the field and the introductory knowledge necessary to succeed in college-level public health coursework. More emphasis needs to be placed on introductory public health concepts before teaching research methods.
 - Additionally, rather than teaching high school students how to conduct research studies, a high school curriculum should instead teach how to read and interpret scientific research. Scientific reading and writing are vital skills to be taught at this stage, since students will more likely be writing literature reviews or class research papers than conducting original, independent clinical research.

Public health problems are often described in EPHS in fragmented ways (person, place, time, data), but not in terms of systemic factors like:

- Redlining, industrial pollution, racism, global trade or climate migration.
- Students should be taught to analyze why inequities exist, not just that they exist which is essential for meaningful intervention.

Methodology for Curriculum:

To adapt the Council on Education for Public Health (CEPH) 2024 undergraduate competencies for a high-school audience, Catalyst Academy undertook a





multi-phase process that prioritized developmental appropriateness, higher-order thinking, and interdisciplinary integration across medicine, public health, and digital health.

Competency Mapping and Simplification

We began by translating CEPH's 11 Foundational Domains (SPH/PHP D9; SBP B1) and six Communication & Information Literacy Competencies (SPH/PHP D10; SBP B2) into 21 age-appropriate learning objectives. Each objective was rewritten at an 8th–10th grade reading level and anchored to a single Bloom's Taxonomy verb (e.g., Understand, Apply, Evaluate, Synthesize). This ensured clarity while fostering cognitive engagement within the Understand → Apply → Analyze band of Bloom's hierarchy.

Catalyst-Specific Competency Integration

To reflect the Academy's interdisciplinary mission, nine additional competencies were introduced across 7 domains: Equity, Critical Thinking, Digital Health, Eight Dimensions of Wellness, Patient Communication, Clinical Assessment, and Interprofessional Development. These align with CEPH's cross-cutting expectations and allow students to explore public health through a broader, more personalized lens.

Instructional Design and Interdisciplinary Integration

Each learning objective was mapped to one of the Academy's three core pillars—Medicine, Public Health, or Digital Health—and reinforced through a corresponding performance task. Tasks included graph creation in Excel, TikTok-style PSAs, policy pitch presentations, literature credibility checklists, and data visualization projects. Lessons were co-developed and co-taught by interdisciplinary teams consisting of clinicians, public health faculty, and digital health practitioners, ensuring that students encountered each concept from multiple professional perspectives.

Performance-Based Assessment

All learning objectives will be paired with an aligned performance task, scored using a three-tiered analytic rubric (Emerging / Proficient / Advanced). These tasks were administered in both pre- and post-program formats to measure individual and cohort-wide growth across cognitive, communication, and applied skill domains.

Expert Review and Iterative Refinement

Drafted competencies, tasks, and rubrics will be reviewed by Public Health faculty, a high school science teacher, a digital health professional, and a student focus



group. Feedback was used to ensure developmental appropriateness, conceptual clarity, and content validity across all instructional components.

Developing Digital Health and AI Competencies for Catalyst Academy

The Catalyst Academy digital health competencies were developed using a structured, top-down methodology designed to adapt professional digital health expectations to a high school learning context. The process began by anchoring the curriculum to the DECODE framework, a widely recognized professional standard for digital health education in healthcare and public health. Rather than adopting DECODE domains directly, the framework was used to identify the core conceptual expectations that learners are assumed to possess before entering formal professional or technical training. This ensured alignment with the field while avoiding premature exposure to advanced system use or technical implementation.

Next, the DECODE domains were decomposed into their foundational conceptual elements. This step focused on identifying the essential ideas that underlie responsible engagement with digital health, including understanding where digital tools are used in healthcare and public health, recognizing health data as both a technical and ethical asset, and appreciating how algorithmic systems influence decision making and equity. These concepts represent the intellectual groundwork that precedes skill acquisition and professional execution.

The identified concepts were then filtered through a developmental appropriateness lens tailored to Catalyst Academy's learner profile. Priority was given to foundational literacy, systems-level understanding, and ethical reasoning, rather than technical proficiency or operational competence. Concepts requiring professional authority, infrastructure access, or advanced quantitative skills were intentionally excluded to ensure that competencies reflected what high school students should understand at this stage, not what trained professionals are expected to perform.

Finally, the remaining concepts were organized into a small set of coherent, Catalyst-specific domains that reflect how digital health functions in real-world contexts. These domains emphasize digital health foundations, health data, privacy and trust, and the role of artificial intelligence in shaping equity and health outcomes. Each competency was written using consistent "understand" language to align with the broader Catalyst curriculum and to reinforce the

program's emphasis on conceptual clarity, ethical awareness, and population-level thinking.

Developing Medical Competencies for Catalyst Academy

The majority of high school students applying to and getting accepted into Project UNITY's Catalyst Academy desire to become involved in healthcare – whether that be physicians, nurses, physician assistants, dentists, and many more healthcare professionals. Although Catalyst Academy is predominantly a public health program, the intersection between medicine, digital health, and public health is an essential requirement to be well-rounded physicians. Therefore, medical competencies were created for Catalyst Academy to start training these young professionals on how to become not just strong public health practitioners, but also astute and effective clinicians.

The medical competencies were developed from the Association of American Medical Colleges's (AAMC) 13 Core Entrustable Professional Activities (EPAs) for Entering Residency. AAMC defines their EPAs as responsibilities/activities that medical students and general health professional trainees must achieve before they graduate from medical school and enter residency. The EPAs are tangible objectives that healthcare students must fulfill upon residency, and represent the daily activities that residents and attendings must do. Therefore, the EPAs were chosen to also provide high school students with insight on the exact roles and responsibilities of clinicians.

The intricacies of all 13 EPAs were read carefully to understand if they have a role in our virtual public health program. Out of 13 EPAs, 4 were chosen to integrate into Catalyst Academy. These EPAs were then transformed into Catalyst Academy Core Competencies 5–9, adapted to fit the high school / early college level.

- EPA 1: Gather a history and perform a physical examination
- EPA 6: Provide an oral presentation of a clinical encounter.
- EPA 9: Collaborate as a member of an interprofessional team.
- EPA 13: Identify system failures and contribute to a culture of safety and improvement.

Catalyst Academy Core Competencies

CEPH-Based High School Core Competencies	
CEPH Domains	Project UNITY's proposed Core Competencies for High School Public Health Curricula
CEPH 1. History of Public Health as a Discipline & Practice	PUPH 1. Understand the history and foundational principles of public health as a field and its role in shaping population-level well-being across time.
CEPH 2. Social Justice & Health Equity as Foundational Public Health Principles	PUPH 2. Understand the drivers of systemic health inequity through an intersectional lens. Understand how social justice is an important tool to advancing health equity sustainably & ethically and how fairness and equal access to care are key values in public health.
CEPH 3. Determinants of Health	PUPH 3. Identify the social, environmental, and behavioral determinants of health. Describe how these factors interact to influence health outcomes and contribute to health disparities.
CEPH 4. Concepts and Applications of Public Health Ethics	PUPH 4. Apply principles of public health ethics to real-world scenarios, including respect for autonomy, justice, beneficence, and transparency.
CEPH 6. Scientific Foundations of Chronic & Infectious Disease	PUPH 5. Explain the biological and environmental causes of chronic and infectious diseases, how they spread within and across populations, and prevention strategies to mitigate spread.
CEPH 7. Fundamental Characteristics of the U.S. Healthcare and Public Health Systems in	PUPH 6. Understand the fundamental characteristics and organizational structures of the U.S. healthcare system. Recognize key differences between systems in other countries. Understand how public health systems impact access and equity.



Comparison with other Nations	
CEPH 9. Public Health Statistical Literacy	PUPH 7. Interpret and communicate basic public health statistics, including risk ratios, prevalence, and trends using charts, graphs, or summaries.
CEPH 10. Public Health Data Collection & Surveillance	PUPH 8. Describe common public health data collection and surveillance methods, including limitations like bias, missing data, and measurement challenges. Explain why evidence-based approaches are essential to practice.
CEPH 11. Planning, Implementing, and Evaluating Evidence-Based Interventions	PUPH 9. Plan and evaluate evidence-based public health interventions, using needs assessments, logic models, and evaluation frameworks.
CEPH 12. Strategies to Build Partnerships & Relationships to Improve Health	PUPH 10. Demonstrate strategies for building partnerships and engaging communities, including collaboration with local government organizations, NGOs, and local stakeholders.
CEPH 13. Application of Critical Thinking Skills to Define and Address Problems in Public Health Practice	PUPH 11. Critically analyze public health problems, evaluate sources of information, and propose actionable, data-informed solutions including a dissemination plan
CEPH 14. Legislative & Governmental Processes Relevant to Public Health Policy and Advocacy	PUPH 12. Describe the role of legislation and policy-making in public health, and how advocacy efforts can influence health outcomes at the local and national levels.



CEPH 15. Policy Analysis	PUPH 13. Evaluate public health policies by assessing their feasibility, ethical considerations, and potential population-level impact.
-	PUPH 14. Understand the concepts of population health and the basic processes, approaches, and interventions used to identify and address major health needs and concerns of populations.
-	PUPH 15. Explore potential career pathways in public health. Describe the importance of youth voices, engagement, advocacy, & involvement in shaping the future of healthcare.
Catalyst Academy-Specific Competencies	
Domain	Catalyst Academy Core Competency
Digital Health Foundations	CACC 1. Understand what digital health is and how digital health tools (e.g., telehealth, wearables, remote monitoring, health apps) are used to support individual and population health.
Health Data, Privacy, and Trust	CACC 2. Understand how health data is collected and used in digital health, and why privacy, consent, transparency, and trust are essential to ethical healthcare and public health practice.
Artificial Intelligence and Equity in Health	CACC 3. Understand how artificial intelligence is used in healthcare and public health, including how bias can be introduced and how AI systems can influence equity, access, and health outcomes.
Holistic Wellness	CACC 4. Identify the eight dimensions of holistic wellness. Describe how they interact and influence one another..





	Understand how each dimension contributes to overall health and well-being.
Patient Communication	CACC 5. Ask effective questions to collect a patient's social determinants of health (SDOH) as part of their social history. CACC 6. List the OLDCAARTS framework for collecting a patient's current history of present illness (HPI).
Clinical Assessment	CACC 7. Summarize a patient's history of present illness based on their symptoms.
Interprofessional Development	CACC 8. Recognize the multidisciplinary team that works to treat patients. CACC 9. Demonstrate how the healthcare system impacts patient care, diagnosis, and treatment.
CEPH Foundational Competencies	
1. Locate public health information	
2. Evaluate public health information	
3. Communicate public health information orally	
4. Communicate public health information in writing	
5. Communicate public health information to a non-specialist audience through a medium other than standard narrative writing: e.g., social media posts, videos, PSAs, brochures, blogs, podcasts, etc.	
6. Synthesize public health information: i.e., compile information from multiple sources and distill it into cohesive conclusions	





Weekly Activities

Topics Covered	Competency	Guest Speakers
Week 1: Introductions to Public Health, Epidemiology, & Biostatistics		
Introduction to Public Health: • The definition & philosophy of public health • The differences between population health, public health, and individual health • The difference between health inequities and health disparities • The difference between community health and population health • Key historical milestones in public health • The difference between health differences and health disparities • The role of public health and medicine in addressing health disparities • Social Determinants of Health (SDOH) • Socio-Ecological Model • Health Belief Model	PUPH 1 PUPH 3 PUPH 5 PUPH 7 PUPH 8 PUPH 11 PUPH 14	
Introductions to Epidemiology & Biostatistics: • The definition of epidemiology and its role in public health • The difference between infectious and chronic diseases • Modes of transmission (e.g., airborne, contact, vector-borne) • Outbreak mitigation methods (e.g., contact tracing, vaccination campaigns) • Key public health statistics (e.g., incidence, prevalence, risk ratio) • Disease trends & comparisons using data visualization (e.g., epidemic curves, bar graphs, pie charts) • Public health surveillance & data collection methods		





• Evidence-based decision-making • Challenges with using data (e.g., bias, missing data)		
Week 2: Health Equity, Community Health, & Community Engagement		
Health Equity & Community Health: • Drivers of health inequities within the healthcare systems (e.g., implicit bias from providers, cost of care, language/cultural barriers, underrepresentation in clinical trials) • Drivers of health inequities outside of the healthcare system (e.g., structural racism, residential segregation, discriminatory policies) • The role of grassroots efforts (e.g., vaccine drives, community education) in addressing health inequities	PUPH 2 PUPH 3 PUPH 9 PUPH 10 PUPH 14	
Community Engagement: • The difference between community service/volunteering and community engagement • Ethical, sustainable, consensual, & collaborative community engagement • Needs assessments, logic models, and evaluation frameworks • Asset-Based Community Development • Savior Complex • The role of personal identities in community engagement • The role of social justice & community engagement work in advancing health equity • The role of youth voices & advocacy in making progress • Health equity tourism		





Importance of research dissemination		
Week 3: Public Health Communication & Misinformation		
- The purpose of public health communication - Elements of effective public health messages (e.g., clarity, accuracy, audience, tone, and cultural relevance) - Examples of effective, trustworthy public health messages - Evidence-based evaluation criteria on the credibility of health information sources - Examples of difference between public health misinformation and disinformation - The role of various channels (e.g., algorithms, social media platforms) in amplifying & spreading public health misinformation - Consequences of health misinformation - Types of bias	PUPH 11 CEPH FC 1 CEPH FC 2 CEPH FC 3 CEPH FC 4 CEPH FC 5 CEPH FC 6	
Week 4: <u>Health Policy, Law, & Ethics</u>		
- Core principles of public health ethics (e.g., justice, autonomy, beneficence, transparency) - The difference between health policy and law - The difference between local, state, and national-level health policies and laws - Examples of public health policies that have improved health (e.g., seatbelt laws, clean water regulations, smoke-free zones) - The role of health policies & laws (e.g., ACA) in shaping healthcare systems and public health - The role of health policies & laws in contributing to health equity - The role of advocacy in influencing policy - Evidence-based policy recommendations	PUPH 4 PUPH 11 PUPH 12 PUPH 13 CEPH 1 CEPH 2 CEPH 3 CEPH 4 CEPH 6	





• Writing evidence-based policy briefs		
Week 5: Digital Health & Artificial Intelligence		
Digital Health: • The definition of digital health and its role in improving individual and population health. • The role of digital health tools (e.g., telehealth, wearables, remote monitoring, health apps) in prevention, chronic disease management, and population health initiatives. • The role of digital health in addressing the Quintuple Aim of healthcare. • The role of equity, access, and user-centered design in developing and implementing digital health solutions. • How health data is generated, collected, and used within digital health systems. • The importance of privacy, consent, transparency, and trust in digital health and public health practice. Artificial Intelligence: • The definition of artificial intelligence and how it is developed and applied in healthcare and public health (e.g., algorithms, machine learning, large language models, natural language processing). • The role of artificial intelligence in healthcare systems (e.g., clinical decision support, imaging, chatbots, workflow automation). • How bias can be introduced into AI systems through data, design, and deployment decisions.	PUPH 8 CACC 1 CACC 2 CACC 3	



• The implications of biased AI systems on equity, access, and health outcomes. • Principles of responsible and ethical use of artificial intelligence in healthcare and public health. • The potential of artificial intelligence to improve health outcomes and advance health equity, as well as its limitations.		
Week 6: <u>Environmental Health, Disaster Management, & Emergency Preparedness</u>		
• The definition of environmental health • The role of environmental factors in influencing individual, community, and population health outcomes • Environmental health risks & their disproportionate impact on vulnerable populations • Climate change & examples of its disproportionate impact on vulnerable populations • The definitions of disaster management & emergency preparedness and how they impact environmental health • The impact of climate change on the frequency and severity of disasters • The role of public health systems (e.g., the CDC, WHO) in preparing and responding to both natural and man-made disasters • The importance of equity in disaster response	PUPH 3 PUPH 11 PUPH 12 PUPH 14	
Week 7: Global Health & Comparative Health Systems		
• The definition of global health • Examples of health challenges and solutions that cross national borders • Social, economic, political, and geographic factors that drive global health inequities • The poverty cycle: how economic inequities	PUPH 3 PUPH 5 PUPH 6	





create global health inequities and vice versa • International organizations and their role in promoting global health • Features and characteristics of the United States healthcare system • Similarities & differences between healthcare systems of different countries and the outcomes they produce	PUPH 10	
Week 8: <u>Career Pathways & the Future of Healthcare</u>		
• The intersection of medicine, public health, and digital health, and its evolving role in solving future healthcare challenges • Careers pathways in medicine, public health, and digital health, and how young people can start preparing for a healthcare career • The importance of young people in shaping the future of healthcare	PUPH 15	

Medical Competencies:

AAMC Competency	PUPH 15	
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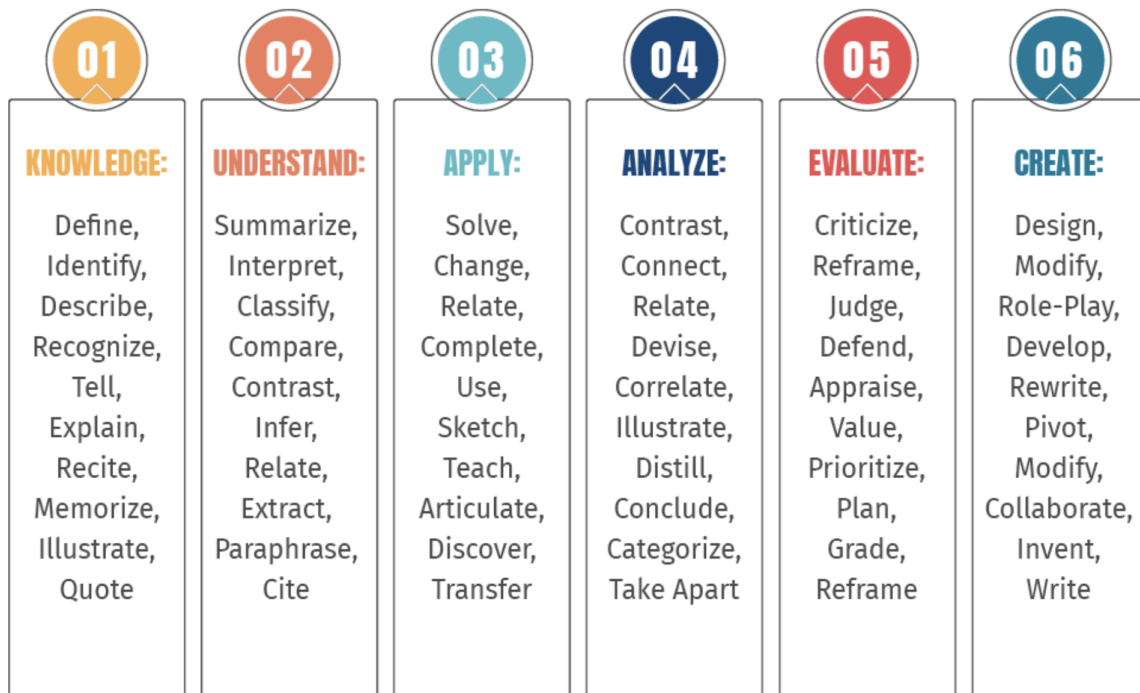
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AAMC most relevant EPAs:

1. Gather a history and perform a physical examination.
 - a. **Ask effective questions to collect a patient's social determinants of health (SDOH) as part of their social history.**
 - b. **List the OLDCAARTS framework for collecting a patient's current history of present illness (HPI).**
2. Provide an oral presentation of a clinical encounter.
 - a. **Summarize a patient's medical and social history for the medical team.**
3. Collaborate as a member of an interprofessional team.



- a. **Recognize the multidisciplinary team that works to treat patients.**
4. Identify system failures and contribute to a culture of safety and improvement
 - a. **Demonstrate how the healthcare system impacts patient care, diagnosis, and treatment.**



Catalyst Wellness Challenge:

To equip students with lifelong wellness skills and help them find fulfillment beyond academics, Catalyst Academy will incorporate a holistic wellness challenge throughout the program. Each week of Catalyst Academy will integrate activities that support one of these eight dimensions, helping students build sustainable wellness habits for long-term success.

8 Dimensions of Wellness:

1. **Physical Wellness**
2. **Emotional Wellness**
3. **Social Wellness**
4. **Intellectual Wellness**
5. **Occupational Wellness**





6. **Spiritual Wellness**
7. **Environmental Wellness**
8. **Financial Wellness**

Career Exploration Assignment:

Designed to help students explore career pathways in healthcare, public health, and AI, while developing critical thinking, research, and presentation skills. Students will investigate a career of interest, analyze its current landscape, and explore how emerging technologies, public health, and policy changes are shaping its future.

Public Health in Real Life Assignment:

Designed to help students identify and analyze real-world public health issues while sharpening their research and communication skills. Students will find and present a current public health topic that has been in the news within the past two weeks, demonstrating its relevance and impact on communities.

One-on-One Mentor Career Pathway Guidance

Designed to provide students with personalized career mentorship, allowing them to explore career pathways, set academic and professional goals, and develop a roadmap for their future. Students will independently research their chosen career field, create a Career Roadmap, and then meet one-on-one with a mentor for feedback and guidance.

Voices of Impact: Igniting Change through Public Health Blogging

Designed to help students develop critical thinking, research, and communication skills by exploring emerging public health issues through blogging. By writing two well-researched and engaging blog posts, students will learn to analyze real-world public health challenges, assess ongoing interventions, and propose actionable solutions.

